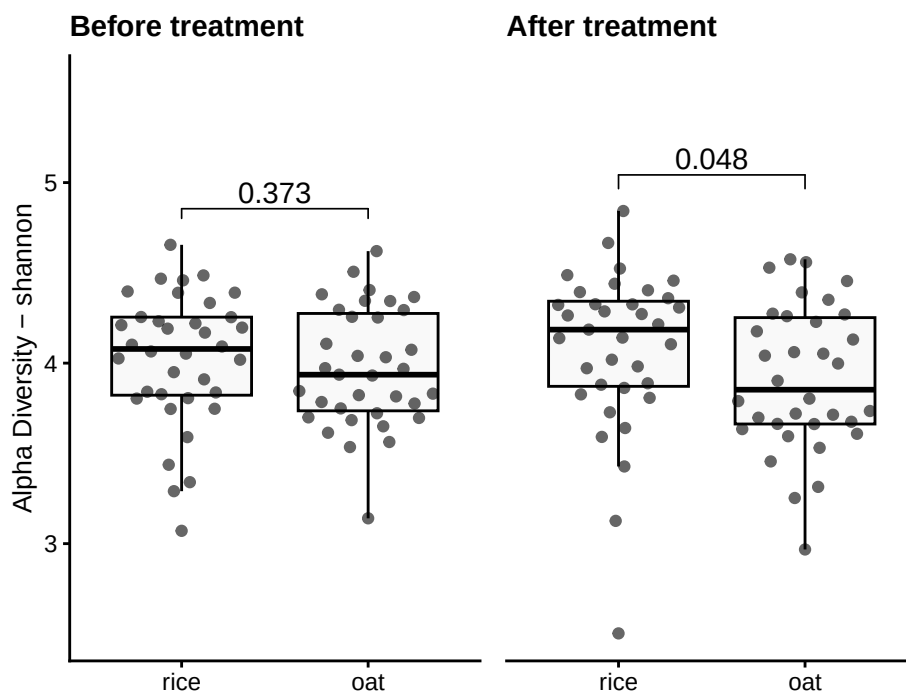


Alpha Diversity

Muluh

1 Group-wise comparisons

- **Index:** shannon
- **Group variable:** diet
- **Multiple testing correction:** FDR
- **Statistical test** Wilcoxon



1.1 Table of the top findings

1.1.1 Across-group

1.1.1.1 Before treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

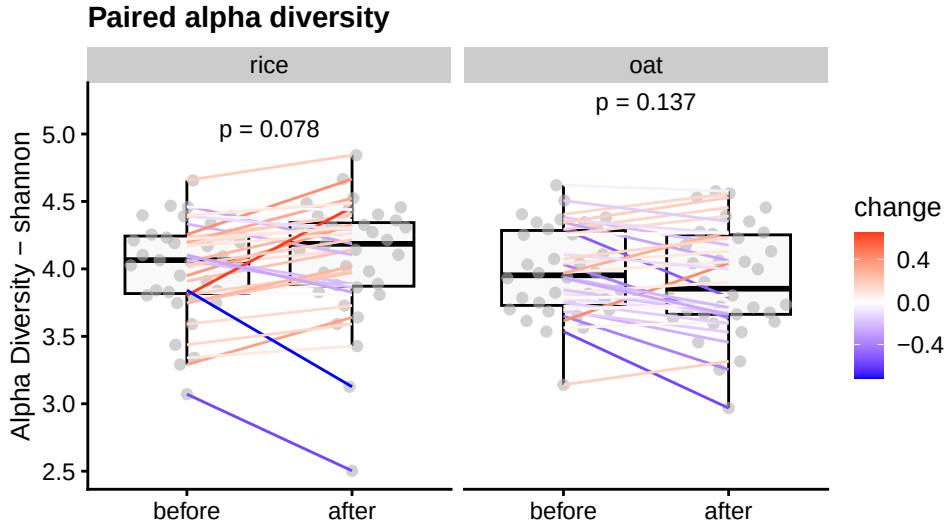
group1	group2	mean1	mean2	log2FC	p.adj
rice	oat	4	4	-0.02	0.373

1.1.1.2 After treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
rice	oat	4.1	3.9	-0.06	0.048



1.1.2 Within-group

1.1.2.1 Paired treatment

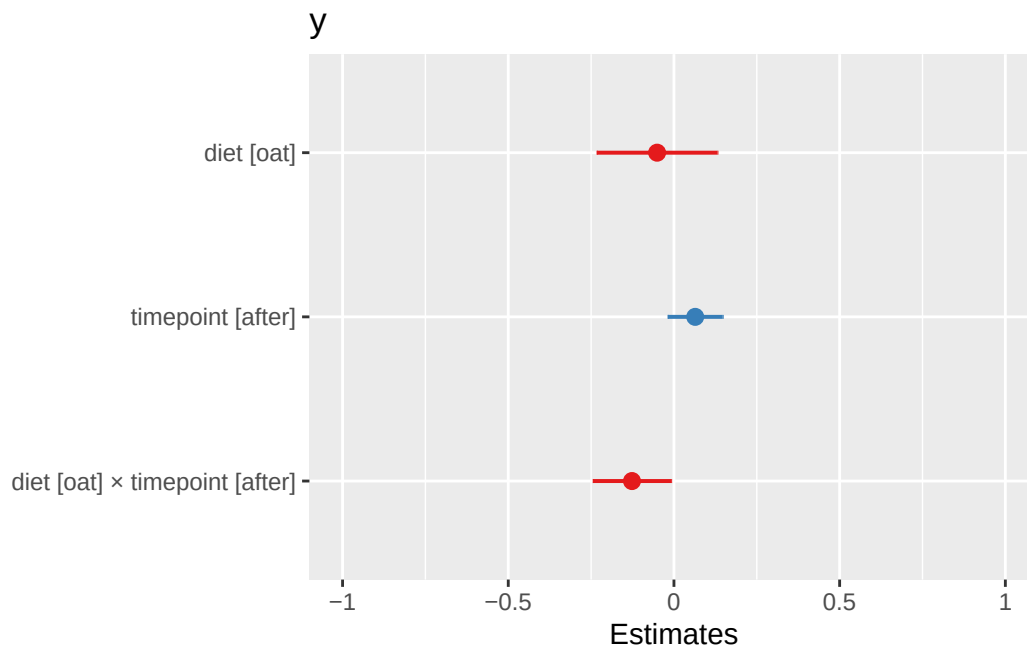
- Multiple testing correction: **FDR**
- Statistical test: **paired Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

diet	mean1	mean2	log2FC	p.value	y.position
rice	4	4.1	0.03	0.078	5.043
oat	4	3.9	-0.02	0.137	5.243

1.2 LMM

- Multiple testing correction: **FDR**
- Statistical test: **paired Wilcoxon**
- Formula used: $\sim \text{diet} * \text{timepoint} + (1 | \text{id})$



2 Data Summary

```

class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(1): relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AE2332 ... VK1336 VK2336
colData names(16): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(20): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```